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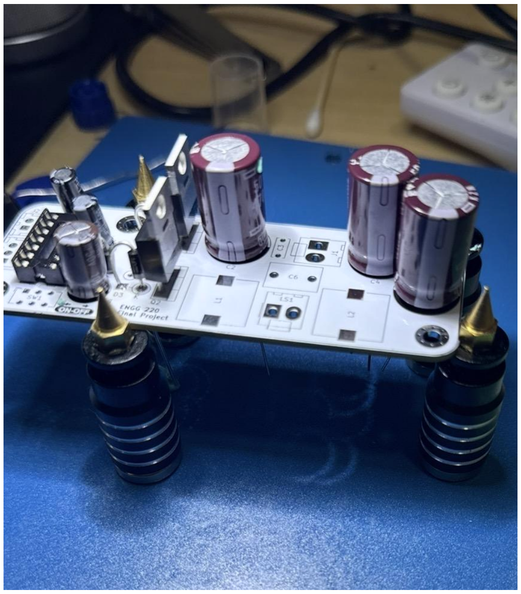
College of Marin

ENGG220 Final Project

05/11/2026

Project Overview

For my final project I built a high-efficiency Class D Amplifier based on the Mundo Electronica design. This project required me to dive deeper into audio filtering and generation of pulse width modulation signals. In addition, I needed to calculate input impedances, components values, and bode plots using LTSpice software . Also, this project required me to read and understand data sheets for integrated circuits.



Pulse Width Modulation (PWM)

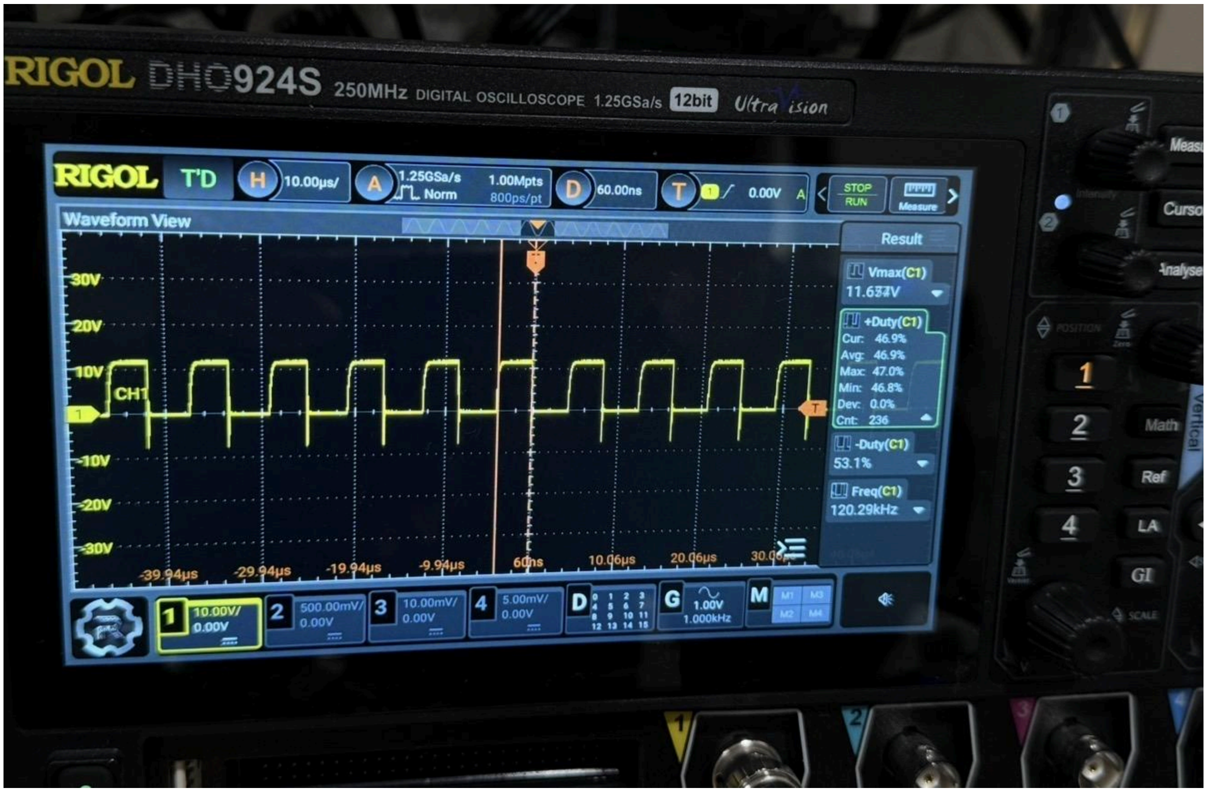
Based on the TL494 (IC) we can generate a sawtooth signal that oscillates between 0 and 3V, which then is compared to the audio signal to generate a PWM signal. I am using a single ended approach which means I have to ground one of the emitters (E2) to get an inverse PWM. This PWM and INV_PWM will be used by the power stage and controlled by IR2110 (IC) which will then allow N-Channel MOSFETS (IRFZ44N's) to increase the power output to the speaker connected. I am implementing a half-bridge topology and using a single source, which means I need a “bootstrap” circuit using IR2110 and a capacitor to give the mosfet a higher voltage than 36V (circuit max) to successfully open and close the gates.

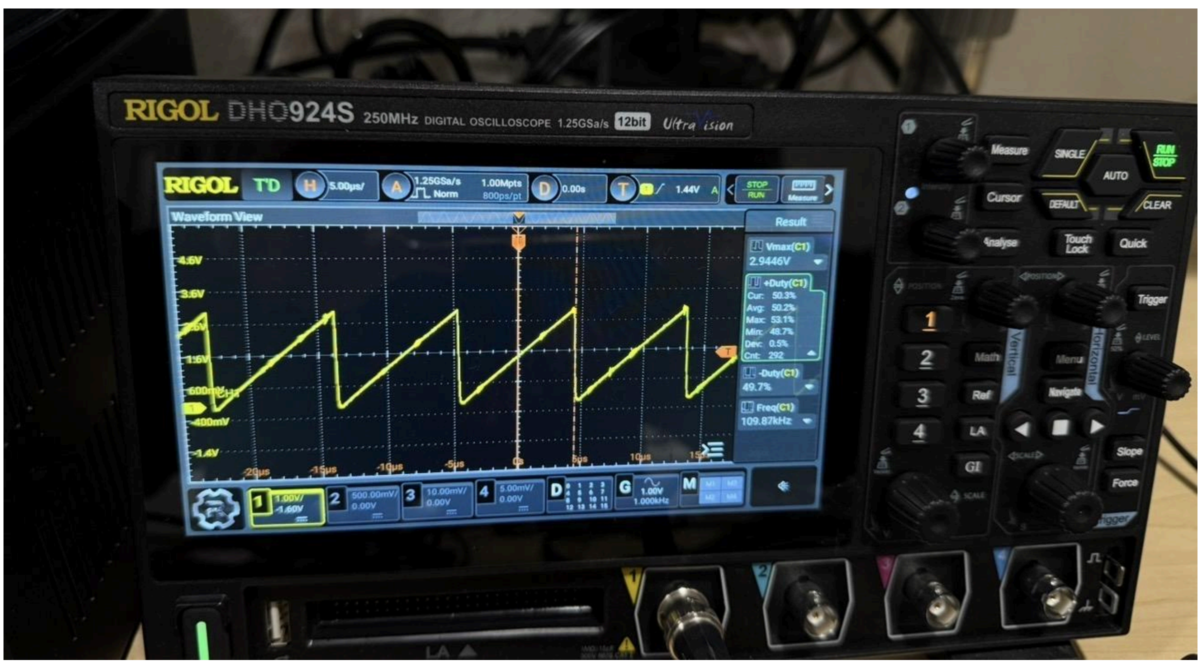
Instructions

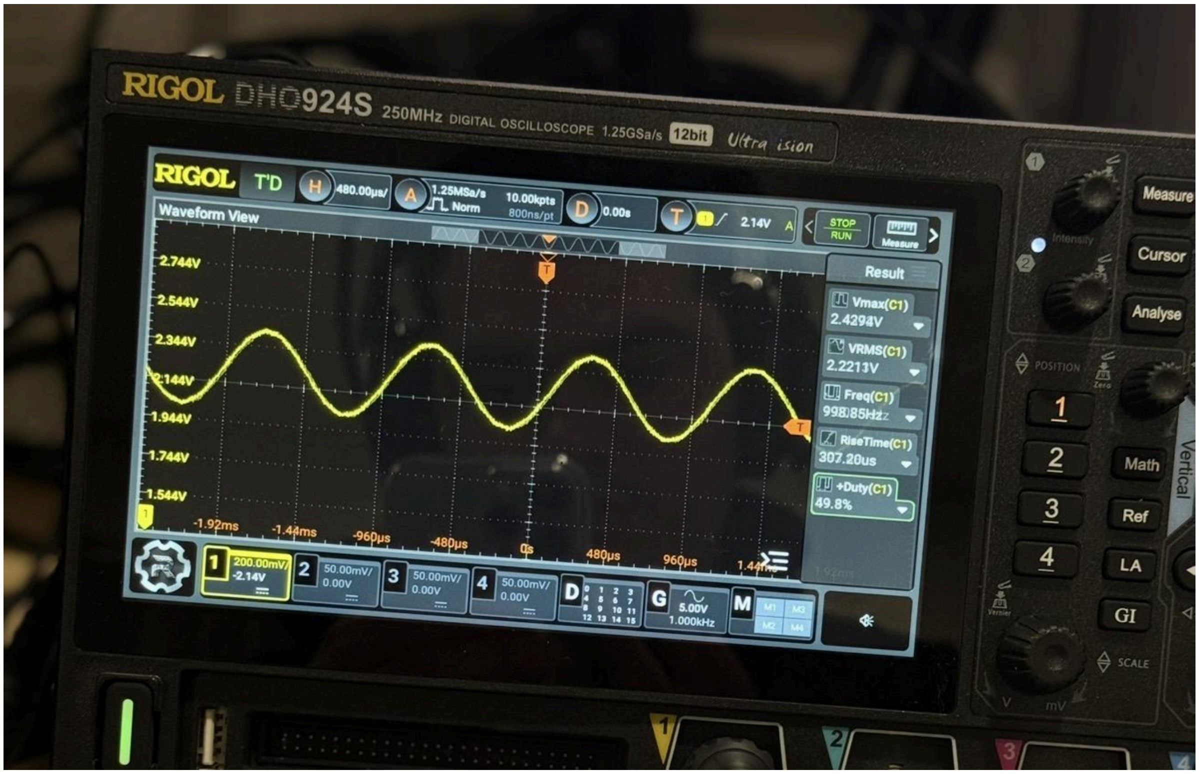
Given that the sawtooth generated by the TL494 oscillates between 0 and 3V we need to "submerge" or audio signal between this range. Now, there are a few ways on how to do this, but the easiest way will be to send a 1khz signal at full volume and measure with an oscilloscope the peak to peak voltage coming out of test point 7 (AUDIO). I have placed a potentiometer that will allow us to adjust the gain. When the gain is too high or above the limits of TL494 or the operational amplifier (LM358) you will experience "clipping" which I have attached a picture for you to see (and avoid).

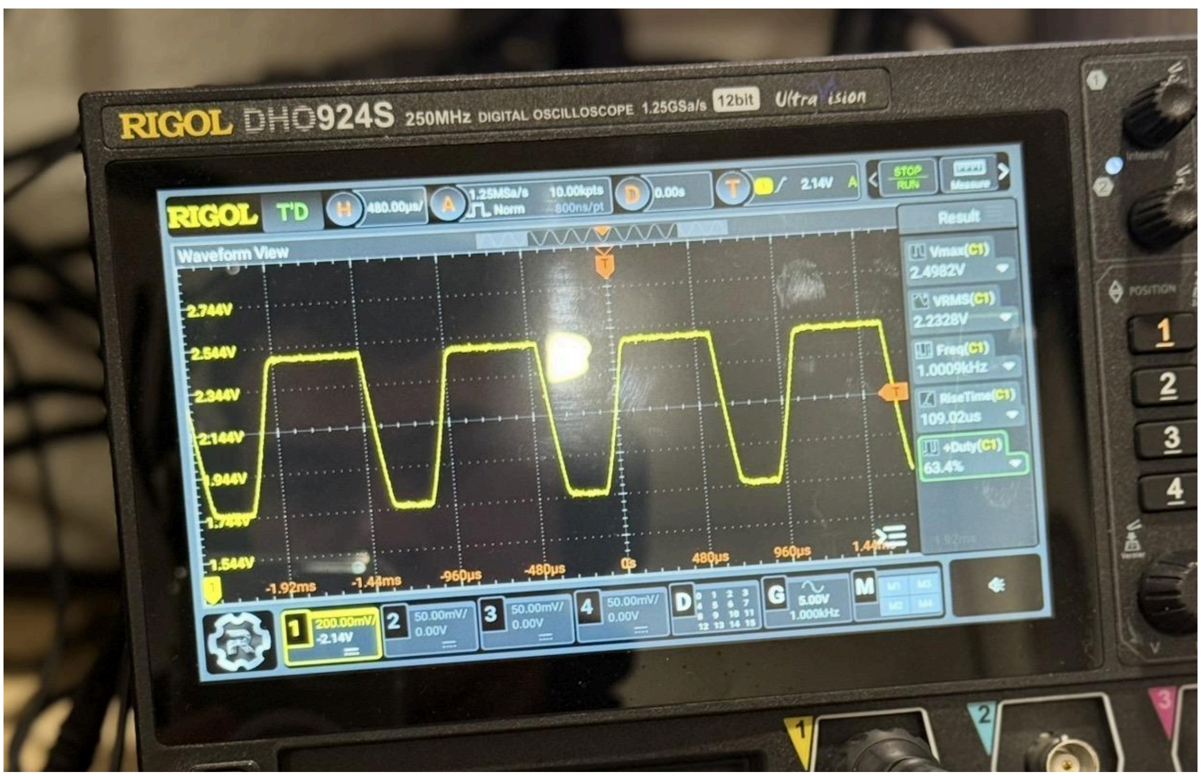
Challenges

I used and calculated resistors that were giving me 0.1 gain for the second operational amplifier, as well as incorrect component values for high pass filters. I blew a couple of transistors and other components. Also, the audio sine wave was not oscillating between 0 and 3V it was oscillating between 2.2V and 2.6V with a peak to peak voltage of only about 400mV. I connected the IC backwards so I needed to double check pin by pin. Soldering challenges and ordering components was also a bit of a struggle and expensive.


RIGOL DHO924S 250MHz DIGITAL OSCILLOSCOPE 1.25GSa/s 12bit
Waveform View
Result
+Duty(C1)
-Duty(C1)
53.1%
Freq(C1)
120.29kHz


RIGOL DHO924S 250MHz DIGITAL OSCILLOSCOPE 1.25GSa/s 12bit
Waveform View
Result
Vmax(C1)
2.9446V
+Duty(C1)
-Duty(C1)
49.7%
Freq(C1)
109.87kHz


RIGOL DHO924S 250MHz DIGITAL OSCILLOSCOPE 1.25GSa/s 12bit
Waveform View
480.00µs/
1.25MSa/s
10.00kpts
Norm
800ns/pt
0.00s
2.14V
STOP RUN
Measure
Result
Vmax(C1)
VRMS(C1)
2.2213V
Freq(C1)
RiseTime(C1)
307.20us
+Duty(C1)
49.8%
2.744V
2.544V
2.344V
2.144V
1.944V
1.744V
1.544V
-1.92ms
-1.44ms
-960µs
-480µs
0s
480µs
960µs
200.00mV/
-2.14V
50.00mV/
0.00V
5.00V
1.000kHz
Cursor
Analyse
Math
Ref
LA
POSITION
SCALE
Vertical


RIGOL DHO924S 250MHz DIGITAL OSCILLOSCOPE 1.25GSa/s 12bit
Waveform View
480.00µs/
1.25MSa/s
10.00kpts
Norm
800ns/pt
0.00s
2.14V
Result
Vmax(C1)
2.4982V
VRMS(C1)
2.2328V
Freq(C1)
1.0009kHz
RiseTime(C1)
109.02us
+Duty(C1)
63.4%
2.744V
2.544V
2.344V
2.144V
1.944V
1.544V
-1.92ms
-1.44ms
-960µs
-480µs
0s
480µs
960µs
200.00mV/
-2.14V
50.00mV/
0.00V
5.00V
1.000kHz